🟢 BASIC LEVEL (1–10)
✅ 1. Print numbers from 1 to N
🧠 Approach

Start loop from 1 to n
Print each number

💻 Code

```
n = 10

for i in range(1, n + 1):
    print(i)
```

✅ 2. Print numbers from N to 1
🧠 Approach

Loop from n down to 1 using step -1

```
n = 10

for i in range(n, 0, -1):
    print(i)
```

✅ 3. Find Sum of First N Numbers
🧠 Approach

Initialize total = 0
Add i inside loop

```
n = 10
total = 0

for i in range(1, n + 1):
    total += i

print("Sum:", total)
```

✅ 4. Find Factorial
🧠 Approach

Multiply numbers from 1 to n

```
n = 5
fact = 1

for i in range(1, n + 1):
    fact *= i

print("Factorial:", fact)
```

☑ 5. Reverse a Number
🧠 Approach

Extract last digit using %
Build reverse

```
n = 1234
rev = 0

while n > 0:
    digit = n % 10
    rev = rev * 10 + digit
    n //= 10

print("Reverse:", rev)
```

☑ 6. Count Digits

```
n = 12345
count = 0

while n > 0:
    count += 1
    n //= 10

print("Digits:", count)
```

☑ 7. Sum of Digits

```
n = 123
total = 0

while n > 0:
    total += n % 10
    n //= 10

print("Sum:", total)
```

☑ 8. Check Palindrome Number

```
n = 121
temp = n
rev = 0

while n > 0:
    rev = rev * 10 + n % 10
    n //= 10

if temp == rev:
    print("Palindrome")
else:
    print("Not Palindrome")
```

✅ 9. Print Even Numbers 1–100

```
for i in range(1, 101):
    if i % 2 == 0:
        print(i)
```

✅ 10. Check Prime Number

```
n = 7
prime = True

for i in range(2, n):
    if n % i == 0:
        prime = False
        break

print("Prime" if prime and n > 1 else "Not Prime")
```

🟡 INTERMEDIATE LEVEL (11–20)

✅ 11. Fibonacci Series

```
n = 7
a, b = 0, 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

✅ 12. Armstrong Number

```
n = 153
temp = n
sum_val = 0

while n > 0:
    digit = n % 10
    sum_val += digit ** 3
    n //= 10

print("Armstrong" if sum_val == temp else "Not Armstrong")
```

✅ 13. GCD of Two Numbers

```
a, b = 12, 18
gcd = 1

for i in range(1, min(a, b) + 1):
    if a % i == 0 and b % i == 0:
        gcd = i

print("GCD:", gcd)
```

✅ 14. LCM of Two Numbers

```
a, b = 12, 18
max_val = max(a, b)
```

```python
while True:
    if max_val % a == 0 and max_val % b == 0:
        print("LCM:", max_val)
        break
    max_val += 1
```

☑ 15. Multiplication Table

```python
n = 5

for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

☑ 16. Count Vowels in String

```python
s = "education"
count = 0

for ch in s:
    if ch in "aeiou":
        count += 1

print("Vowels:", count)
```

☑ 17. Reverse String

```python
s = "hello"
rev = ""

for ch in s:
    rev = ch + rev

print(rev)
```

☑ 18. Print Star Pyramid

```python
n = 4

for i in range(1, n + 1):
    for j in range(n - i):
        print(" ", end="")
    for j in range(2*i - 1):
        print("*", end="")
    print()
```

☑ 19. Inverted Pyramid

```python
n = 4

for i in range(n, 0, -1):
    print(" "*(n-i) + "*"*(2*i-1))
```

☑ 20. Floyd’s Triangle

```python
n = 4
```

```
num = 1

for i in range(1, n + 1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

🟣 HARD LEVEL (21–31)

☑ 21. Diamond Pattern

```
n = 3

for i in range(1, n + 1):
    print(" "*(n-i) + "*"*(2*i-1))

for i in range(n-1, 0, -1):
    print(" "*(n-i) + "*"*(2*i-1))
```

☑ 22. Hollow Square

```
n = 5

for i in range(n):
    for j in range(n):
        if i==0 or i==n-1 or j==0 or j==n-1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

☑ 23. Perfect Number

```
n = 28
sum_val = 0

for i in range(1, n):
    if n % i == 0:
        sum_val += i

print("Perfect" if sum_val == n else "Not Perfect")
```

☑ 24. Strong Number

```
n = 145
temp = n
sum_val = 0

while n > 0:
    digit = n % 10
    fact = 1
    for i in range(1, digit + 1):
        fact *= i
    sum_val += fact
```

```python
    n //= 10

print("Strong" if sum_val == temp else "Not Strong")
```

✅ 25. Pascal Triangle

```python
n = 5

for i in range(n):
    num = 1
    for j in range(i + 1):
        print(num, end=" ")
        num = num * (i - j) // (j + 1)
    print()
```

✅ 26. Count Prime Numbers in Range

```python
count = 0

for num in range(1, 101):
    prime = True
    if num < 2:
        prime = False
    for i in range(2, num):
        if num % i == 0:
            prime = False
            break
    if prime:
        count += 1

print("Total Primes:", count)
```

✅ 27. Sum of Series (1 + 1/2 + 1/3 ...)

```python
n = 5
total = 0

for i in range(1, n + 1):
    total += 1/i

print(total)
```

✅ 28. Find Largest Digit in Number

```python
n = 4829
largest = 0

while n > 0:
    digit = n % 10
    if digit > largest:
        largest = digit
    n //= 10

print("Largest Digit:", largest)
```

✅ 29. Print Multiplication Table 1–10

```
for i in range(1, 11):
    for j in range(1, 11):
        print(i*j, end="\t")
    print()
```

✅ 30. Check Automorphic Number

```
n = 25
square = n * n

if str(square).endswith(str(n)):
    print("Automorphic")
else:
    print("Not Automorphic")
```

✅ 31. Count Frequency of Each Digit

```
n = 122333
freq = {}

while n > 0:
    digit = n % 10
    if digit in freq:
        freq[digit] += 1
    else:
        freq[digit] = 1
    n //= 10

print(freq)
```